

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, ...
Regular & Supplementary Summer – 2024

Semester: IV

Course: B. Tech

Branch: Electronics and Telecommunication Engineering/ Electronics and
Communication Engineering/ Electronics Engineering

Subject Code & Name: BTBS404 Probability Theory and Random Processes

Duration: 3 Hr.

Max Marks: 60

Date: 20/06/2024

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) What is random experiment? Give the relative frequency definition of probability with an example.	(L1/CO1)	6
B) State and prove Bayes theorem.	(L1/CO3)	6
C) A fair coin is tossed 4 times. Define the sample space corresponding to this random experiment. Also find the following probabilities.	(L3/CO1)	6
a) More heads than tails are obtained.		
b) Tails occur on the even numbered tosses		
Q.2 Solve Any Two of the following.		12
A) Show that:	(L1/CO4)	6
a) $E\{aX + bY\} = aE\{X\} + bE\{Y\}$		
b) $Var\{aX + bY\} = a^2 var\{X\} + b^2 var\{Y\} + 2ab c\{X, Y\}$		
where $c\{X, Y\}$ is covariance of X and Y		
B) Explain probability density function (pdf) and cumulative distribution function (cdf) along with their properties for one continuous and discrete random variable.	(L2/CO1)	6
C) Define independent events and prove following theorems related to independent events.	(L1/CO3)	6
a) $P(A \cap B) = P(A) \cdot P(B)$		
b) $P(\bar{A} \cap B) = P(\bar{A}) \cdot P(B)$		

- Q. 3 Solve Any Two of the following.** **12**
- A) Define and explain Poisson distribution and prove its Mean and Variance. (L1/CO2) **6**
 - B) State and prove Schwarz Inequality. (L1/CO4) **6**
 - C) If 3% of electronic units manufactured by a company are defective. Find the probability that in a sample of 200 units, less than 2 bulbs are defective. (L3/CO1) **6**
- Q.4 Solve Any Two of the following.** **12**
- A) State and explain Almost Sure Convergence. (L1/CO3) **6**
 - B) Write a note on law of large numbers and explain the types. (L2/CO3) **6**
 - C) Define convergence. What are the different types of convergence. State the relation between them. (L2/CO3) **6**
- Q. 5 Solve Any Two of the following.** **12**
- A) If $\{X(t)\}$ is WSS process with autocorrelation $R(\tau) = Ae^{-\alpha|\tau|}$. Determine the second order moment of RV $X(8) - X(5)$. (L3/CO2) **6**
 - B) State Autocorrelation function related to stationary process and prove any two properties. (L1/CO2) **6**
 - C) Define random process. provide the classification of random process based on nature of space and time parameter along with example for each type. (L2/CO2) **6**

***** End *****